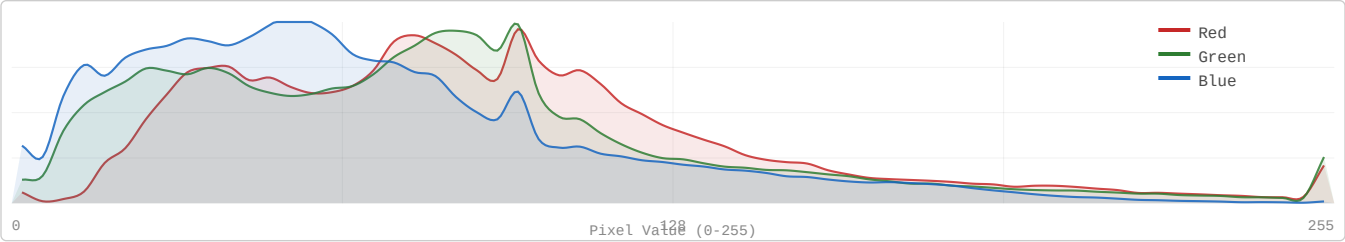
Decorrelation Gap: KODIM05

768x512 | 24-bit RGB | PNG (lossless)

Autumn Stream with Mossy Rocks | Weakly one-dimensional (PC1 = 91.91%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.8981	0.8215
G	0.8981	1.0000	0.9034
B	0.8215	0.9034	1.0000

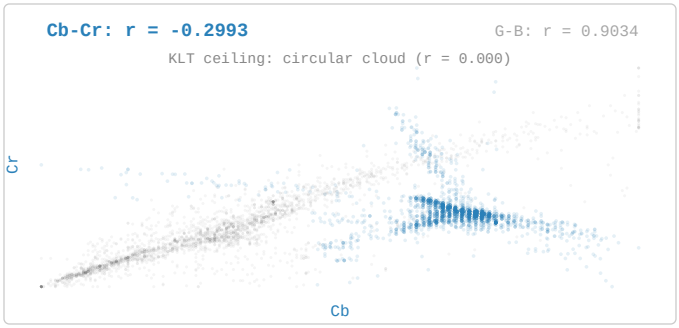
Cb-Cr: -0.299

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	-0.3854	-0.1219
Cb	-0.3854	1.0000	-0.2993
Cr	-0.1219	-0.2993	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.898130	-0.385368	0.3854
R-B → Y-Cr	0.821497	-0.121895	0.1219
G-B → Cb-Cr	0.903419	-0.299333	0.2993
Avg |r|	0.874349	0.268865	0.2689

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5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

69.2% High

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

11.4%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	41.56 dB
G	46.35 dB
B	38.99 dB
Overall	41.36 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

PC1	91.91%
PC2	5.59%
PC3	2.51%

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5794	-0.6251	-0.5230	6478.81
PC2	-0.7287	0.1100	0.6759	394.03
PC3	0.3651	-0.7727	0.5193	176.61

8. PCA Derived Metrics

Condition Number

36.68

PC1/PC2: 16.44 | PC2/PC3: 2.23

Blue Channel Independence

11.4%

Variance not captured by PC1

PC1 Dominant Channel

G

Loading: G (0.63)

9. Redundancy Profile

Classification: Weakly one-dimensional | CN: 36.68 | PC1: 91.9%

RGB Correlation: Avg |r| = 0.874349 | R-G: 0.8981 | R-B: 0.8215 | G-B: 0.9034

YCbCr Residual: Avg |r| = 0.268865 | Y-Cb: -0.3854 | Y-Cr: -0.1219 | Cb-Cr: -0.2993

Efficiency: 69.2% of KLT ceiling | Gap: 0.2689 avg |r| remaining

Blue Channel: 11.4% independent | 4:2:0 PSNR: 41.36 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.